# Lecture 7 — Nyquist Stability Criterion

The Nyquist criterion relates the open-loop frequency response to the number of closed-loop poles in the right half plane. A system is stable when the Nyquist plot encircles the point −1 exactly P times counter-clockwise.

Gain margin and phase margin are read from the Bode diagram. Typical design targets are a phase margin above 45 degrees and a gain margin above 6 dB.

Homework: problems 9.4, 9.7 and 9.12 — due Thursday.